

Heng-I Chu

(765) 772-6283 | ivorchu@gmail.com | [linkedin.com/in/ivorchu](https://www.linkedin.com/in/ivorchu) | <https://ivorchu.github.io>

Education

Carnegie Mellon University

M.S. in Electrical and Computer Engineering

Pittsburgh, PA

Dec 2027

Purdue University

B.S. in Computer Engineering

West Lafayette, IN

May 2026

Concentration in *Computer Systems / Microelectronics*

GPA: 3.8/4.0

Work Experiences

Academia Sinica

Intern Researcher

Taipei, Taiwan

Jun 2025 - Aug 2025

- Designed an end-to-end compiler datapath in AITemplate, lowering PyTorch models to optimized CUDA kernels via NVIDIA CUTLASS, accelerating sparse general matrix multiplication on Ampere GPU Architectures.
- Extended structured sparsity support from 2:4 to general **N:M formats**, designing a runtime compression pipeline, custom CUDA kernels with warp-level parallelism, and optimized memory layouts for high throughput.
- Integrated optimized sparse operations into the compiler stack, involving frontend operator configuration, backend kernel launch tuning, and scheduling strategies, resulting in an average of **1.8 times speedups** over dense kernels and outperforming NVIDIA's default 2:4 implementation in certain configurations.

Industrial Technology Research Institute

AI Software Engineer Intern

Hsinchu, Taiwan

Jun 2024 - Aug 2024

- Engineered an end-to-end self-driving pipeline integrating ROS, solid-state LiDARs, and 3D point cloud detection and tracking models, achieving 3 times faster real-time object detection by deploying NVIDIA TensorRT and ONNX format parsing.
- Implemented multimodal sensor fusion by aligning point cloud and image data with a custom matching algorithm.
- Researched and developed point cloud generative models by concatenating VAE, PointNet++, Diffusion Models, and GANs to produce virtual point-cloud scenes of busy roads and improved Point Cloud Detection model **performances by 15%**.

Projects & Relevant Experiences

Accel-Sim / GPGPU-Sim Hopper & Blackwell GPU Architecture Modeling

Purdue Accelerator Architecture Lab (Prof. Timothy Rogers, Tor Aamodt)

West Lafayette, IN

Aug 2025 - Present

- Extended Accel-Sim / GPGPU-Sim to support NVIDIA Hopper and Blackwell features by adding PTX ISA 8.x/9.x support, including WGMMMA, descriptor-based shared memory, and synchronization primitives with correct warp-group coordination.
- Modified the cycle-level timing model to incorporate warp-group scheduling, tensor core pipeline behavior, and asynchronous execution semantics, including issue/commit behavior, dependency tracking, and descriptor-based memory access patterns
- Collected execution traces using NVBit and validated simulator behavior against real GPU runs with CUTLASS examples.

High-Bandwidth DDR4 Memory Controller

Purdue SoCET AI Hardware

West Lafayette, IN

Aug 2025 - Present

- Built and verified a JEDEC-compliant DDR4 controller (Micron model, 1333 MT/s), including command sequencing, timing, refresh, and data transfer for the Atalla AI accelerator; extended it with a non-blocking AXI-style design and per-bank queues.
- Integrated CMU Ramulator with the RTL DRAM controller via Verilator and DPI-C, implementing a cycle-synchronized wrapper that bridges AXI and scratchpad traffic to the Ramulator's timing model for end-to-end verification.

Compiler for Custom Tensor Core

Purdue SoCET AI Hardware

West Lafayette, IN

Aug 2025 - Present

- Built a compiler backend that translates C code into a custom 64-bit CISC ISA designed for a custom accelerated matrix processor.
- Designed a VLIW (Very Long Instruction Word) packet scheduler with latency-aware dependency tracking, software scoreboard, and register-bank allocation, achieving up to 3.2x higher ILP and 2.5x IPC improvement over prior scoreboard designs.
- Co-designed the Atalla ISA architecture, incorporating APx extensions, vector code generation, loop unrolling, and memory layout tuning; reducing core-logic area by 1.8x compared to RISC-V extension on the MITLL 90 nm process.

Multicore Pipelined Processor System

Purdue University

West Lafayette, IN

Aug 2025 - Dec 2025

- Designed and implemented multi-cycle, pipelined, and dual-core RISC-V processors with full hazard detection, data forwarding, branch prediction, and integrated set-associative L1 cache, validated on the Urbana FPGA with variable-latency memory.
- Built cache and memory-controller state machines implementing arbitration, cache-to-cache transfers, snooping, writeback, LR/SC support, and full MSI coherence to enforce correct shared-memory behavior across cores.

Skills & Relevant Courses

- Programming: C, C++, Python, JAVA, SystemVerilog, CUDA, Assembly (RISC-V, x86), ROS
- Software Systems: LLVM Compilers, Operating Systems, AITemplate, PyTorch, TensorFlow, Keras, CUTLASS, TensorRT, ONNX
- Hardware Design: GPU architecture, DDR4 DRAM, AXI, cache coherence, FPGA, Verilator, DPI-C, GPGPU-Sim, Ramulator